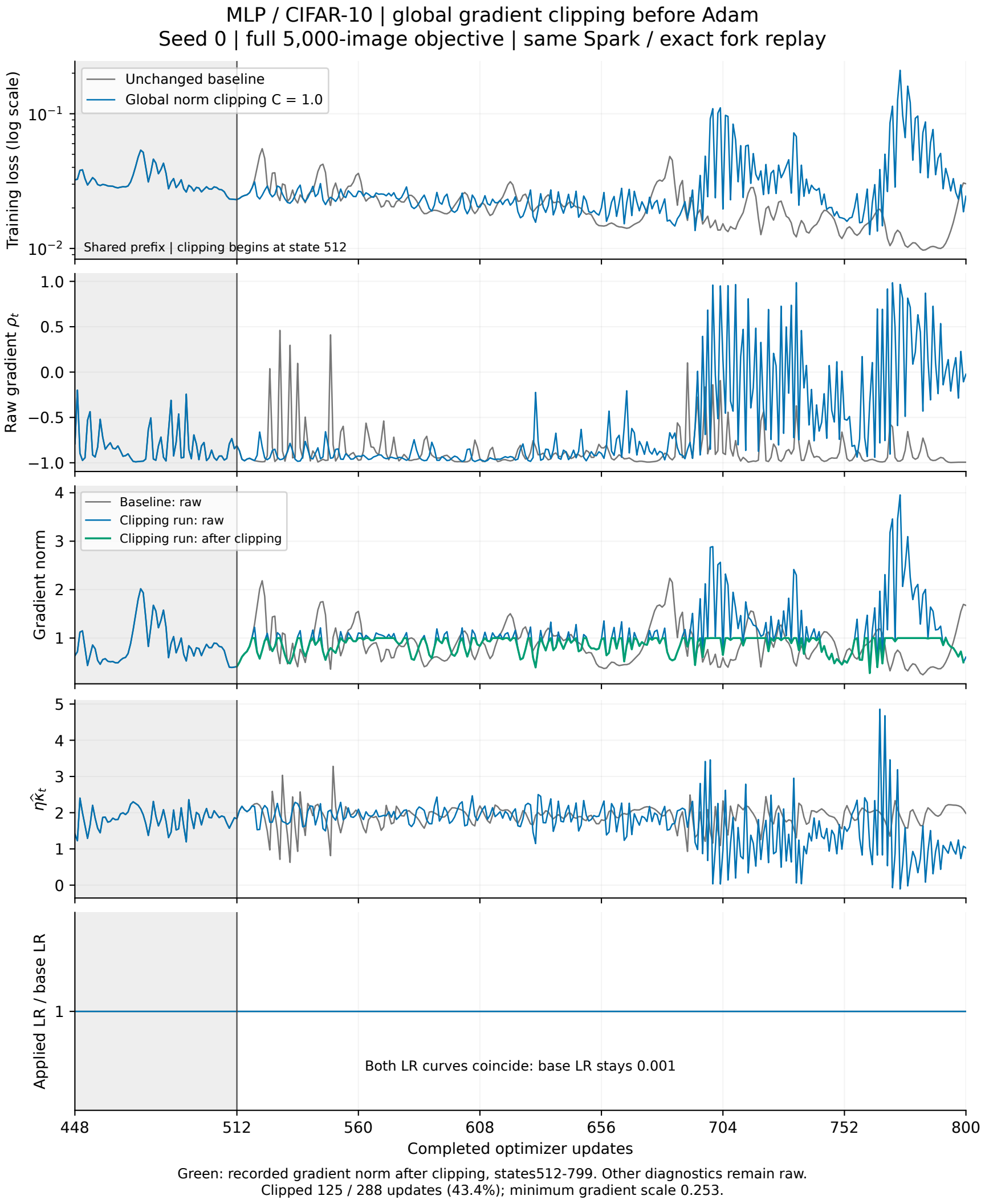

MLP / CIFAR-10 | global gradient clipping before Adam
Seed 0 | full 5,000-image objective | same Spark / exact fork replay
Training loss (log scale)
Unchanged baseline
Global norm clipping C = 1.0
Shared prefix | clipping begins at state 512
Raw gradient ρ_t
Gradient norm
Baseline: raw
Clipping run: raw
Clipping run: after clipping
$\eta\hat{\kappa}_t$
Applied LR / base LR
Both LR curves coincide: base LR stays 0.001
Completed optimizer updates
Green: recorded gradient norm after clipping, states512-799. Other diagnostics remain raw.
Clipped 125 / 288 updates (43.4%); minimum gradient scale 0.253.